

Ecoskiller – COMPLETE LIST

Docker – Package applications and dependencies into portable containers

Kubernetes – Orchestrate, scale, self-heal, and deploy containerized services

Helm – Versioned and repeatable Kubernetes deployments

NGINX – Ingress routing, TLS termination, traffic control

Kong – Central API gateway for routing, auth enforcement, rate limits

Envoy – High-performance rate limiting, retries, circuit breaking

ModSecurity – Protect APIs from OWASP Top-10 web attacks

Keycloak – Authentication, OAuth, SSO, MFA, RBAC

HashiCorp Vault – Secure storage of secrets, API keys, credentials

Open Policy Agent – Centralized authorization and policy-as-code enforcement

PostgreSQL – Primary transactional database with ACID guarantees

Redis – Caching, locks, timers, GD orchestration state

OpenSearch – Full-text search for jobs, candidates, recruiters

ClickHouse – High-volume analytics, metrics, scoring data

Flyway – Version-controlled database schema migrations

MinIO – Store resumes, certificates, invoices, audit files

etcd – Strong-consistency distributed locking and coordination

Jitsi – Voice-only group discussions with forced mute/unmute control

LiveKit – Scalable real-time audio/video for interviews and dojo

coturn – NAT traversal for reliable WebRTC connectivity

Apache Kafka – Event bus for decoupled, event-driven microservices

RabbitMQ – Background jobs and async task processing

Apache Airflow – Scheduled workflows for billing, analytics, reports

Postfix – Outgoing email delivery engine

Docker Mail Server – Self-hosted complete email server bundle

Jasmin SMS Gateway – SMS delivery for OTPs and notifications

ntfy – Lightweight push notifications

Prometheus – Metrics collection and alerting

Grafana – Dashboards for metrics, analytics, system health

Loki – Centralized log storage and querying

OpenTelemetry – Distributed tracing and telemetry

Wazuh – SIEM, intrusion detection, security audits

Velero – Backup and restore Kubernetes clusters and data

Terraform – Provision reproducible cloud infrastructure

GitHub Actions – Automated build, test, and deploy pipelines

GitLab CI – Alternative CI/CD pipeline engine

Unleash – Runtime feature toggles and tenant-based rollout

I. SYSTEM SHAPE (NON-NEGOTIABLE)

Architecture style

- Event-driven microservices
- Strict separation: **business logic** ≠ **realtime** ≠ **media**
- Stateless services + stateful infra
- Multi-tenant from day zero

Traffic classes

1. HTTP APIs (CRUD, queries)
 2. WebSocket commands (GD, interviews, dojo matches)
 3. Media (WebRTC/Jitsi/LiveKit – never through your API)
 4. Async events (analytics, scoring, billing)
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II. CORE INFRASTRUCTURE SERVICES

1. Container & Orchestration

- Docker (all services)
- Kubernetes (mandatory)
- Ingress Controller (NGINX or Traefik)
- Internal Service Mesh (optional at scale)

2. Compute Layout (k8s namespaces)

- core – auth, users, jobs
 - realtime – GD, interviews, dojo
 - media – Jitsi / LiveKit
 - billing
 - analytics
 - admin
 - ops
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III. DATA LAYER (BY PURPOSE)

Primary Databases

- PostgreSQL

- users, recruiters, jobs, applications
 - sessions, interviews, dojo matches
 - billing, subscriptions, invoices
- Row-level security for tenant isolation

In-Memory / State

- **Redis**
 - GD state machines
 - interview timers
 - rate limiting
 - OTPs
 - locks (slot booking)

Search

- **OpenSearch / Elasticsearch**
 - candidate discovery
 - job search
 - recruiter filters

Analytics

- **ClickHouse**
 - GD metrics
 - speaking time
 - scoring distributions
 - dojo performance
 - funnel analytics

IV. MEDIA & REALTIME STACK (LOCKED)

Voice / Video

- **Self-hosted Jitsi** (voice GD, recruiter-less)
- **LiveKit (SFU)** for dojo matches & interviews
- **coturn** for TURN/STUN

Rules:

- Media never touches backend
- Backend only issues **short-lived tokens**

- Rooms = session_id / match_id
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V. CORE BACKEND MICROSERVICES (MANDATORY)

1. Auth Service

Handles:

- email/password
 - OAuth
 - JWT access + refresh
 - MFA
 - device sessions
 - RBAC
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2. User Service

- candidate profiles
 - recruiter profiles
 - mentor profiles
 - tenant binding
 - skill vectors
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3. Recruiter Service

- company profiles
 - recruiter approval
 - job posting limits
 - billing entitlements
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4. Job Service

- job CRUD
- job categories
- salary bands
- moderation flags

5. Application Service

- job applications
- pipeline stages
- rejection reasons
- recruiter actions

6. Interview Service

- interview scheduling
- slot locking
- token generation
- reminders
- interview lifecycle

7. Voice GD Orchestrator (CRITICAL)

This replaces humans.

Responsibilities:

- batch creation
- room naming
- join window enforcement
- turn-based speaking engine
- forced mute/unmute
- failure handling
- numeric data capture

Tech:

- Node.js / Spring Boot
- Redis deterministic state machine
- WebSocket command channel

8. Dojo Match Engine

- match creation
 - scenario assignment
 - role binding
 - timer enforcement
 - state transitions
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9. Scoring Engine (LOCKED)

- weighted metrics
 - peer + mentor merge
 - variance detection
 - immutable audit logs
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10. Certification & Belt Engine

- eligibility checks
 - mentor confirmation
 - certificate generation
 - versioned belts
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11. Analytics Service

- consumes event bus
 - writes to ClickHouse
 - exposes dashboards
 - anomaly detection
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12. Notification Service

- email
 - SMS
 - push
 - templates
 - orchestration
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13. Billing & Subscription Service

- plans
 - feature gating middleware
 - usage metering
 - invoices
 - refunds
 - GST/VAT
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14. Admin Governance Service

- moderation
 - disputes
 - mentor misconduct
 - audit review
 - platform config
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15. Integration Hub

- LinkedIn
 - GitHub
 - LMS / HRIS (future)
 - Webhook framework
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VI. ASYNC & EVENTING

Event Bus

- **Kafka or RabbitMQ**

Events:

- user.created
- job.applied
- interview.completed
- gd.completed
- match.scored
- belt.eligible

- invoice.generated

No synchronous chaining across domains.

VII. DEVOPS PIPELINE (LOCKED)

CI

- GitHub Actions / GitLab CI
- build
- unit tests
- security scans
- image push

CD

- Helm charts
- blue/green deploy
- automated rollback
- versioned DB migrations

Environments

- dev
- staging
- prod

No manual prod deploys.

VIII. OBSERVABILITY (NON-OPTIONAL)

Metrics

- Prometheus

Logs

- Loki or ELK

Tracing

- OpenTelemetry

Dashboards

- GD failure rate
 - interview no-shows
 - scoring anomalies
 - billing errors
 - media QoS
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IX. SECURITY BASELINE

- WAF in front of ingress
 - rate limits per IP + user
 - signed media tokens
 - encrypted PII
 - immutable audit logs
 - consent capture for recordings
 - tenant isolation enforced at DB
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X. FINAL REALITY CHECK

Approximate components:

- **Microservices:** 15–18
- **Infra services:** 12+
- **Media stack:** 3–4
- **Pipelines & ops:** 6+

Total moving parts: **~40–45**